

PRML PROJECT

Title: Facial Feature Analysis for BMI Prediction and
Offence Profiling in Inmate Populations

INTRODUCTION

- Overview of the project
 - Predict gender and BMI using facial images (frontal and side views).
 - Applications in healthcare, biometric systems, and offender profiling.
- Objectives:
 - Gender classification from facial features.
 - BMI estimation and grading (underweight, normal, overweight).

DATASET DESCRIPTION

- Dataset Details:
 - Images: Frontal and side-view facial images.
 - Attributes: Gender, BMI, and demographic details.
- Preprocessing:
 - Images resized to 224×224, normalized, and converted to RGB format.
 - Data split: 80% training, 16% validation, 4% testing.

FEATURE ENGINEERING

- Some of the Extracted Features for BMI Estimation(total 13 features considered):
 - Face length-to-width ratio.
 - Lip-to-chin distance.
 - Neck-to-face ratio.
 - Double chin features.
 - Cheekbone prominence.
 - Jawline sharpness.
 - Eye spacing and symmetry.
 - Forehead-to-face proportion.
- Insights from Correlation Heatmap:
 - Significant relationships between features and BMI.

FEATURE ENGINEERING RESULTS

- Feature Correlation Heatmap helps us select features with higher correlation with bmi or gender for training the models
- perceptron, SVM and random forest trained but poor accuracy and R^2 score because of low correlation between the features and the bmi/gender
- As Machine learning approach on the fails because of the nature of dataset and feature engineering limitations we switch to deep learning model
- RSNet 50 model improves the performance significantly

NEURAL NETWORK ARCHITECTURE

- Base Model:
 - ResNet50 pre-trained on ImageNet, up to global average pooling.
- Branch Networks:
 - Separate dense layers for frontal and side-view image processing.
- Fusion Layer:
 - Concatenation of features from both branches.
- Output Layers:
 - BMI prediction (linear activation).

MODEL TRAINING AND OPTIMIZATION

- Loss Functions:
 - BMI: Mean Squared Error (MSE).
 - Gender: Categorical Cross-Entropy.
- Metrics:
 - BMI: R^2 score and Pearson correlation.
- Optimization:
 - Optimizer: Adam (learning rate 0.001).
 - Regularization: Dropout (rate 0.5).
- Training:
 - Batch size: 16.
 - Epochs: 40 (early stopping on validation loss).

RESULTS AND INSIGHTS

- BMI Estimation:
 - R^2 Score: 0.6.
 - Pearson Correlation: 0.6.
 - Features like face width, lip-to-chin distance contribute significantly.
- Fusion Effectiveness:
 - Combined frontal and side views improved results.

CONCLUSION

- Achievements:
 - Successfully integrated ResNet50 for feature extraction.
 - Achieved high accuracy in gender classification (94%).
 - Reliable BMI prediction with R^2 score of 0.6 and correlation of 0.6.
- Real-World Applications:
 - Biometric systems, offender profiling, healthcare assessments.

TABLE

Model Used	RSNET ₅₀	Dataset used	Illionis
Pearson correlation coefficient	0.6	Front / side images used (if both are used, mention as both)	Front
R ² error	0.6	No.of Images in training	52800
MSE	34.41	testing	13200
MAE	9.007	Accuracy (if applicable)	94